

As i mentioned since there is a change in scope, hmi is one of the major deliverable from our end Please look at the screenshot And review , add points if I missed anything Create dedicated deliverable and associated tasks for hmi

After looking at your screenshots and considering the **latest scope change**, I would say the HMI has now become a **major WP5.3 deliverable**, not just a small integration task.

However, several items in the screenshots are still based on the older "Spatial Digital Twin HMI" concept and should be updated.

What should be removed or renamed?

Remove / de-emphasize

✗ Digital Twin wording

Replace:

Spatial Digital Twin HMI
Digital Twin HMI

with:

Process Intelligence HMI
Process Analytics HMI
Process Insight & Visualization HMI

because the MoMs explicitly moved away from a full digital twin.

✗ Root Cause Analysis Framework as a major deliverable

Current screenshot contains:

Advanced Analytics, Process Anomaly Detection & Root Cause Framework

Root cause analysis is still useful, but should not be a primary deliverable.

Rename to:

Process Anomaly Analysis & Engineering Insights Framework

✗ Automated Root Cause Diagnosis

Current:

Deploy automated Root Cause Analysis Framework

Too ambitious for current scope.

Replace with:

Provide engineering investigation support tools

What should be added?

These are now central according to the MoMs.

Process State Visualization

The MoMs repeatedly mention:

Layup
Cut/Add
Retract
Travel
Approach

Add:

Visualize current process state
Visualize process-state transitions
Display process-state timeline

Expected-vs-Actual Visualization

This is one of the key outputs from:

- time-series analysis

- behavioral model

Add:

Expected KPI trend

Actual KPI trend

Deviation visualization

Expected process envelope visualization

KPI Superimposition on Part Geometry

This was explicitly discussed.

Add:

Map KPIs onto 3D part geometry

Map anomalies onto 3D geometry

Map process variations onto 3D geometry

This should be a dedicated task.

Process Activation Visualization

This was mentioned multiple times.

Add:

Visualize process activation sequence

Display executed path timeline

Display process progress along part

Recommended New Deliverable Structure

D11 — HMI Architecture & Visualization Framework Design

Duration

~4 Weeks

Output

- HMI Architecture
 - Visualization Architecture
 - Data Model
 - Frontend/Backend Architecture
 - Screen Definitions
 - UX Wireframes
-

D11.1 Requirements Analysis (1 Week)

Define:

- operator use cases
 - engineering use cases
 - anomaly visualization requirements
 - KPI visualization requirements
 - process-state visualization requirements
 - 3D superimposition requirements
-

D11.2 Architecture Definition (1 Week)

Define:

- frontend architecture
 - backend architecture
 - API architecture
 - event pipeline
 - streaming architecture
 - deployment architecture
-

D11.3 Visualization Design (1 Week)

Design:

- KPI dashboards
 - anomaly dashboards
 - expected-vs-actual views
 - process-state views
 - process activation views
 - engineering analysis views
-

D11.4 Data Model & Interface Definition (1 Week)

Define:

- KPI schema
 - anomaly schema
 - process-state schema
 - geometry mapping schema
 - API payloads
-

D12 — HMI Backend Development

Duration

~6–8 Weeks

Output

- Backend services
 - APIs
 - Data pipelines
 - Geometry mapping services
-

D12.1 Backend Framework Setup (1 Week)

Setup:

- backend stack
- database
- logging

- deployment
-

D12.2 Data Ingestion Services (1.5 Weeks)

Implement:

- KPI ingestion
 - time-series ingestion
 - process-state ingestion
 - metadata ingestion
-

D12.3 Geometry & Spatial Mapping Engine (1.5 Weeks)

Implement:

- CAD/mesh loading
- geometry indexing
- timestamp → XYZ mapping
- KPI → geometry mapping
- anomaly → geometry mapping

This is a major missing item in many plans.

D12.4 Analytics Integration Layer (1 Week)

Integrate:

- anomaly outputs
 - behavioral model outputs
 - expected-vs-actual outputs
-

D12.5 API Development (1 Week)

Implement:

- dashboard APIs
- anomaly APIs
- process-state APIs

- geometry APIs
-

D12.6 Backend Validation (1 Week)

Validate:

- performance
 - synchronization
 - API consistency
-

D13 — HMI Frontend Development

Duration

~8–10 Weeks

Output

- Operator HMI
 - Engineering HMI
 - 3D Visualization
 - Anomaly Visualization
-

D13.1 Frontend Setup (1 Week)

Setup:

- React/Angular/Vue
 - charting framework
 - 3D rendering framework
-

D13.2 KPI Dashboard Development (1.5 Weeks)

Implement:

- KPI trends

- process health
 - machine health
-

D13.3 Process-State & Activation Visualization (1.5 Weeks)

Implement:

- process timeline
- process-state transitions
- execution progress

This is currently missing from your screenshots.

D13.4 3D Geometry Visualization (2 Weeks)

Implement:

- part visualization
 - zoom/pan/selection
 - layer management
-

D13.5 Anomaly & KPI Superimposition (1.5 Weeks)

Implement:

- anomaly markers
- KPI heatmaps
- deviation overlays
- expected-vs-actual overlays

This is one of the most important WP5.3 features.

D13.6 Engineering Analysis Views (1 Week)

Implement:

- anomaly investigation view
- KPI correlation view

- process variation view
-

D13.7 Frontend Validation (1 Week)

Validate:

- performance
 - usability
 - rendering
 - synchronization
-

D14 — End-to-End HMI Validation & Deployment

Duration

~3 Weeks

Output

Production-ready HMI

D14.1 Integration Testing (1 Week)

Validate:

- backend ↔ frontend
 - analytics ↔ HMI
 - geometry ↔ KPI mapping
-

D14.2 User Acceptance Testing (1 Week)

Validate:

- operator workflows

- engineering workflows
-

D14.3 Deployment & Documentation (1 Week)

Deploy:

- HMI stack
 - user guides
 - maintenance guide
-

My recommendation is to treat **Geometry Mapping + KPI/Anomaly Superimposition + Process-State Visualization** as first-class deliverables. Those are the areas most clearly aligned with the latest WP5.3 scope and are not fully captured in the screenshots you shared.

can you give detailed information about tasks instead of bulletpoints?

Yes. For project planning and charter discussions, detailed task descriptions are much better than short bullet points because they explain:

- **why the task exists,**
- **what activities are performed,**
- **what output is expected,**
- **how it contributes to WP5.3 objectives.**

Below is a more detailed version for the HMI-related deliverables.

D11 — HMI Architecture & Visualization Framework Design

Objective

Define the overall HMI architecture that will support process intelligence, anomaly visualization, process activation monitoring, KPI visualization, and integration with the WP5.3 analytics modules. The goal is to establish a scalable architecture before any software development begins.

Expected Output

- HMI architecture document
- Frontend/backend architecture
- Visualization architecture
- Data flow architecture
- Screen definitions and wireframes

Duration

~4 Weeks

D11.1 — HMI Requirement Analysis & Use Case Definition

Duration: ~1 Week

Description

The first activity is to understand how operators, process engineers, and manufacturing engineers will use the system. Since the objective is not simply to display machine values but to support process understanding and anomaly investigation, the team must define what information is required at different user levels.

The task includes identifying which KPIs are important, how anomalies should be represented, how process states should be visualized, and how users should navigate between high-level process views and detailed engineering analysis views.

Expected Output

- User requirements document
 - Use-case catalogue
 - KPI visualization requirements
 - Anomaly visualization requirements
-

D11.2 — HMI System Architecture Definition

Duration: ~1 Week

Description

This activity defines the overall software architecture of the HMI platform. The objective is to separate data processing responsibilities from visualization responsibilities and define how information flows from the analytics engines into the user interface.

The architecture should define how time-series outputs, anomaly indicators, process-state information, and behavioral-model outputs are exchanged between backend services and frontend dashboards.

The deployment architecture should also be defined, including communication protocols, APIs, databases, and interaction with WP5.1 infrastructure.

Expected Output

- HMI architecture diagram
 - Component architecture
 - Communication architecture
 - Deployment architecture
-

D11.3 — Visualization Framework & UX Design

Duration: ~1 Week

Description

The objective of this task is to design the visual experience of the HMI.

This includes defining dashboard layouts, navigation concepts, anomaly views, KPI trend views, process activation views, expected-versus-actual comparison screens, and engineering analysis pages.

Wireframes and mockups should be produced to validate the visualization concept with stakeholders before development begins.

Expected Output

- Dashboard mockups
 - Screen wireframes
 - Navigation flow
 - Visualization concept
-

D11.4 — Data Model & Interface Definition

Duration: ~1 Week

Description

The HMI requires a consistent representation of KPIs, anomalies, process states, geometry information, and analytics outputs.

This task defines the data structures that will be exchanged between the backend and frontend. Standardized interfaces should be defined for anomaly events, KPI trends, process-state transitions, expected-versus-actual metrics, and geometry-based visualization.

Expected Output

- API specification
 - Data schema definitions
 - Integration contracts
-

D12 — HMI Backend Development

Objective

Develop the backend services responsible for ingesting process information, interfacing with analytics modules, mapping anomalies to geometry, and providing data services to the frontend dashboards.

Expected Output

- Backend services
- APIs
- Data integration services
- Geometry mapping services

Duration

~6–8 Weeks

D12.1 — Backend Framework Setup

Duration: ~1 Week

Description

This task establishes the software foundation for backend development. The development framework, database technology, deployment strategy, and communication mechanisms are selected and configured.

Logging, configuration management, and service orchestration are also established to ensure future scalability and maintainability.

Expected Output

- Backend project structure
 - Database connectivity
 - Development environment
-

D12.2 — Data Ingestion & Synchronization Services

Duration: ~1.5 Weeks

Description

The objective of this task is to ingest process data streams and prepare them for visualization.

The backend must be capable of receiving KPI values, process-state information, anomaly indicators, machine status information, and analytics outputs. Synchronization logic is required to ensure that data arriving from multiple sources is aligned correctly in time.

Special attention should be paid to timestamp handling and consistent data representation.

Expected Output

- Data ingestion services
 - Synchronization services
 - Time-aligned process data streams
-

D12.3 — Geometry Mapping & Spatial Visualization Backend

Duration: ~1.5 Weeks

Description

One of the key WP5.3 requirements is the ability to superimpose process information onto part geometry.

This task develops the backend services responsible for loading CAD or mesh representations, maintaining geometry metadata, and mapping KPI values and anomaly events to physical locations on the part.

The backend should provide spatial information that enables frontend visualization of process variations and anomalies directly on the component geometry.

Expected Output

- Geometry management service
 - Spatial mapping engine
 - KPI-to-geometry mapping service
-

D12.4 — Analytics Integration Layer

Duration: ~1 Week

Description

This task integrates outputs from the time-series anomaly detection framework and the behavioral modeling framework.

The backend must collect anomaly events, expected-versus-actual metrics, process-state information, and KPI deviations and transform them into a common format suitable for visualization.

Expected Output

- Analytics integration service
- Unified process-intelligence data model

D12.5 — Backend API Development

Duration: ~1 Week

Description

The backend must expose APIs that allow the frontend to retrieve process information efficiently.

These APIs should support KPI dashboards, anomaly dashboards, geometry overlays, process-state visualization, and expected-versus-actual analysis.

Both real-time and historical queries should be considered.

Expected Output

- REST/WebSocket APIs
- API documentation

D12.6 — Backend Validation & Performance Testing

Duration: ~1 Week

Description

Before frontend integration begins, backend services must be validated.

Testing should verify API correctness, synchronization accuracy, geometry mapping accuracy, data consistency, and overall system performance under realistic workloads.

Expected Output

- Backend validation report
- Performance assessment

D13 — HMI Frontend Development

Objective

Develop the operator-facing and engineering-facing dashboards used to visualize process behavior, anomalies, KPI trends, process activation states, and geometry overlays.

Expected Output

- Complete HMI application
- KPI dashboards
- Anomaly dashboards
- Geometry visualization
- Engineering analysis views

Duration

~8–10 Weeks

D13.1 — Frontend Framework Setup

Duration: ~1 Week

Description

Establish the frontend framework, visualization libraries, state-management architecture, and communication mechanisms required to build the HMI.

This task creates the foundation on which all visualization components will be developed.

Expected Output

- Frontend project structure
 - Visualization framework
-

D13.2 — KPI Dashboard Development

Duration: ~1.5 Weeks

Description

Develop dashboards capable of displaying process KPIs, machine KPIs, process-health indicators, and KPI trends.

The dashboards should support real-time updates and allow users to navigate between different process views.

Expected Output

- KPI dashboards
 - Process-health views
-

D13.3 — Process-State & Process Activation Visualization

Duration: ~1.5 Weeks

Description

The objective is to visualize the progression of manufacturing operations.

Users should be able to see which process state is active, where the process currently is within the execution sequence, and how process-state transitions occur over time.

This becomes particularly valuable during anomaly investigation.

Expected Output

- Process timeline
 - Process-state dashboard
-

D13.4 — 3D Geometry Visualization

Duration: ~2 Weeks

Description

This task develops the interactive 3D visualization environment used to display part geometry and associated process information.

Users should be able to navigate the geometry, inspect locations of interest, and understand where process variations and anomalies occurred.

Expected Output

- Interactive 3D viewer
 - Geometry interaction tools
-

D13.5 — KPI & Anomaly Superimposition

Duration: ~1.5 Weeks

Description

This is one of the most important WP5.3 visualization capabilities.

The frontend should display KPI trends, process variations, and anomaly indicators directly on the geometry. This allows users to understand not only that an anomaly occurred, but also where it occurred and how it relates to the manufacturing process.

Expected Output

- KPI overlays
 - Anomaly overlays
 - Spatial process visualization
-

D13.6 — Engineering Analysis Views

Duration: ~1 Week

Description

Develop dedicated engineering analysis views that allow users to investigate process behavior in greater detail.

These views should provide expected-versus-actual comparisons, KPI deviation analysis, anomaly timelines, and behavioral-model outputs.

Expected Output

- Engineering analysis dashboard
 - Expected-vs-actual visualization
-

D13.7 — Frontend Validation & User Testing

Duration: ~1 Week

Description

Validate the usability, responsiveness, and correctness of the HMI.

Testing should ensure that dashboards update correctly, geometry visualizations remain responsive, and anomaly overlays accurately reflect analytics outputs.

Expected Output

- User acceptance report
 - Frontend validation report
-

This level of detail is typically suitable for a **project charter, work package plan, or detailed roadmap**, because each task explains the purpose, activities, expected outputs, and estimated duration rather than simply listing bullets.